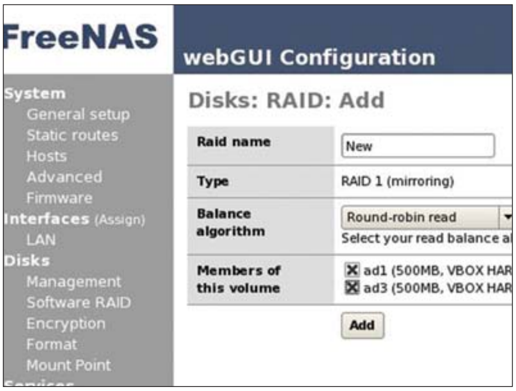
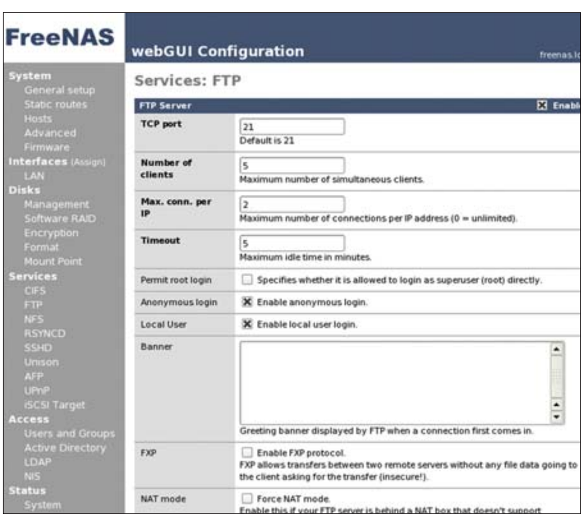


Set up a RAID array to protect your data better

this particular machine is done. You can now administer your FreeNAS server from any workstation on the network, using the Web user interface—just enter the server's IP address in your browser's address bar and hit [**Enter**]. The default username is `admin` and the password is `freenas`—which you should change when you're done with your settings.

The Core

The first thing you need to do is add your hard disks under `Disks > Management`. Use the + icon to begin adding your disks. A hard disk standby time of 30 minutes will give you a respectable balance between performance and power savings if your network is going to be busy; if the NAS box isn't going to be used too often, you might even consider pulling it down to five minutes. You can also decide how much you want to trade off between performance and



Your shared disks can be accessible through various protocols—even FTP

disk noise—very useful if you work in a small, quiet place where disk noise can become extremely bothersome. Leave the Preformatted FS option as Unformatted—you should format your disks under `Disks > Format` when you're done with this part.

If you have two identical hard disks, consider using a RAID-1 setup: the two hard disks will appear as one, and data will be written to both of them simultaneously—so if one disk fails, your data is still intact. The disadvantage, unfortunately, is that you lose the extra storage the second hard disk offers you. To start with your software RAID, select `Software RAID` under `File System` in Disks > Format for both the disks. `Under Disks > Software RAID > RAID-1`, use the + icon to add a new RAID array; give it a name and select the hard disks you want to add. FreeNAS also supports other RAID formats—for more dope on them, refer to *Know More About RAID* in our January 2007 issue.

If you're on a predominantly Windows-based network, the CIFS protocol works best

Spread The Joy

Once you're done setting up your hard disks, it's time to share them with the world. The first thing to do is mount the disks and give them a share name, which will let people access the disk over the network. You'll find this option under `Disks > Mount Point`. The next step is to create Groups and Users (in that order, under `Access > Users and Groups`) who will be authorised to access the shared drives. If you're on a Windows domain, use the Active Directory option—this will automatically authenticate all the people on your network using their domain usernames and passwords.

Finally, it's time to decide on a sharing protocol. If you're on a predominantly Windows-based network, the CIFS protocol works best. Your shared drives will show up as regular Windows shares, and any Linux client that wanders into your network can use a Samba client to connect to it. On a Linux-based network, the Network File System (NFS) is a better option—it's the native distributed file system for Unix- and Linux-based networks.

If you want your server to be accessible over the Internet, enable the FTP protocol as well, but do ensure that you disable Anonymous Logins. You'll also need to configure your router to send FTP requests to the server's IP address.

Beyond

At this point, you have a basic, working NAS server. You can now customise it—its name, the admin password and so on—under `System > General Setup`. You can even encrypt your disks for an added level of security.

When you're through, disable the console menu under `System > Advanced`, and disconnect your keyboard and monitor from the main server. To ensure that nobody removes your USB disk, you could get yourself a separate front-panel USB attachment, but instead of attaching it to your front panel, leave it loose. Connect the drive and then leave inside the system's cabinet.

Endnote

One of the disadvantages of FreeNAS is that its user management is somewhat rudimentary—it's either grant or deny. FreeNAS has none of the depth we're used to with regular operating systems. That aside, if you have old hardware that can't serve a better purpose and don't want to invest in a new server, FreeNAS is the perfect storage solution for your network. ⊠

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